# STAT 3400 - Homework 6

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## Problem 1

There are two common representations of a simple logistic regression model. The first is in the probability form shown below.

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}}$$

The second representation is the log-odds form shown below.

$$\log\left(\frac{P(y = 1|x)}{P(y = 0|x)}\right) = \beta_0 + \beta_1 x$$

Using some algebra, show that these two representations are mathematically equivalent.

**Solution:** Let's start from $\log\left(\frac{P(y=1|x)}{P(y=0|x)}\right) = \beta_0 + \beta_1 x$.

Exponentiate both sides: $\left(\frac{P(y=1|x)}{P(y=0|x)}\right) = e^{\beta_0+\beta_1 x}$.

Since this is binary classification, let $p = P(y = 1|x)$, then $P(y = 0|x) = 1 - p$, so we can write the equation as $\frac{p}{1-p} = e^{\beta_0+\beta_1 x}$.

Solve for p by multiplying both sides by $1 - p$: $p = e^{\beta_0+\beta_1 x}(1 - p) = e^{\beta_0+\beta_1 x} - e^{\beta_0+\beta_1 x}p$.

We can factor out p: $p(1 + e^{\beta_0+\beta_1 x}) = e^{\beta_0+\beta_1 x}$, then we can solve for p: $p = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$.

Finally if we multiply numerator and denominator by $e^{-(\beta_0+\beta_1 x)}$: $p = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$.

Since $p = P(y = 1|x)$, we have $P(y = 1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$.

## Problem 2

This problem regards the conversion between various forms of representing the likelihood of an event. Specifically, the problem involves a binary response variable that is equal to 1 when an individual defaults on their credit card debt and 0 when they do not default.

### Part A

On average, what fraction of people with an odds of 0.37 of defaulting on their credit card debt will in fact default? Show your work.

**Solution:** Let p be the probability that a person defaults. Then, $0.37 = \frac{p}{1-p}$. Which is equivalent to $0.37 * (1-p) = p$, $0.37 = 1.37p$. Thus, $p = \frac{0.37}{1.37} = 0.27....$ Therefore, about 27% of people with odds 0.37 of defaulting will actually default.

### Part B

Suppose that an individual has a 16% chance of defaulting on their credit card debt. What are the odds that the individual will default? Show your work.

**Solution:** In this case, since the $p = 0.16$, the odds are $\frac{0.16}{1-0.16} = \frac{0.16}{0.84} = 0.19...$ Therefore, the odds that individual will default is about 0.19.

## Problem 3

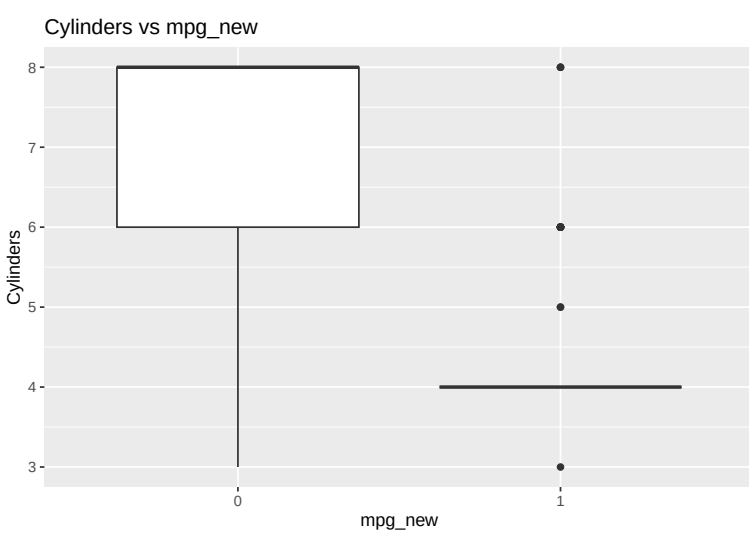
This problem uses the `Auto` data set from the `ISLR2` package.

```
library(ISLR2)
data(Auto)
```

### Part A

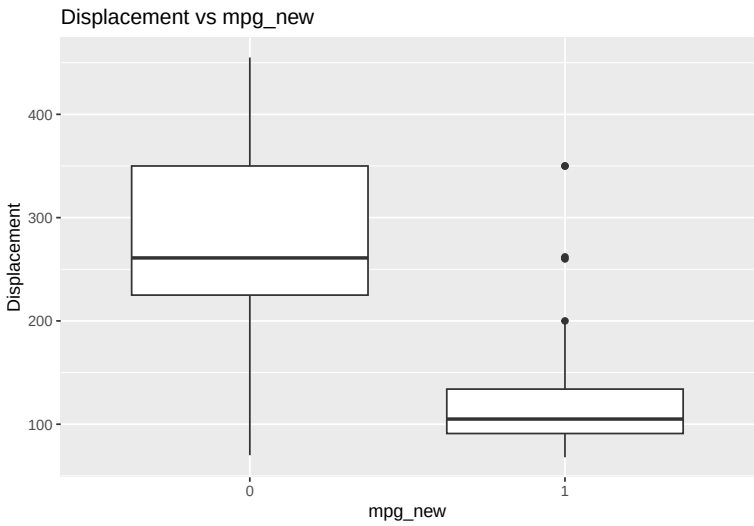
Add a new binary variable `mpg_new` that is equal to 1 when `mpg` has a value greater than its median and 0 otherwise. Then, after removing `year`,`origin`, and `name` from the data set, create five separate box plots that show the association between `mpg_new` and the remaining predictors. Which of the predictors appears to be most associated with `mpg_new` based on what you observe in the plots?

```
mpg_median <- median(Auto$mpg, na.rm = TRUE)
Auto$mpg_new <- ifelse(Auto$mpg > mpg_median, 1, 0)
Auto_2 <- select(Auto, -year, -origin, - name)
ggplot(Auto_2, aes(x = factor(mpg_new), y = cylinders)) +
  geom_boxplot() +
  labs(title = "Cylinders vs mpg_new",
       x = "mpg_new",
       y = "Cylinders")
```

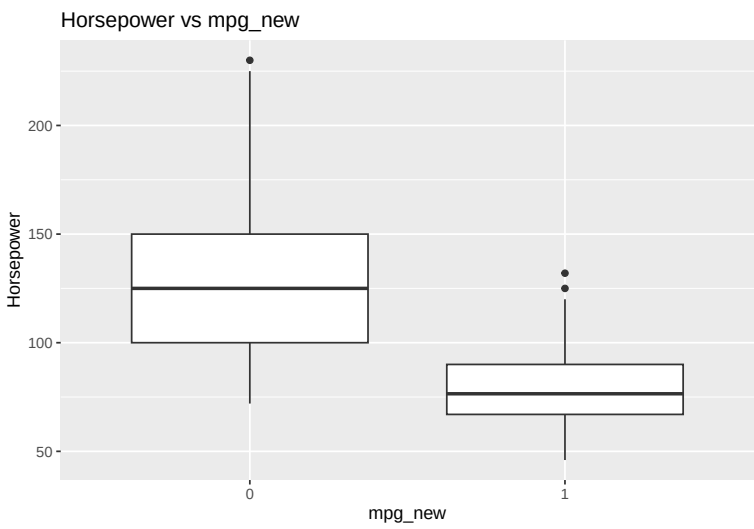



```
ggplot(Auto_2, aes(x = factor(mpg_new), y = displacement)) +
  geom_boxplot() +
  labs(title = "Displacement vs mpg_new",
```

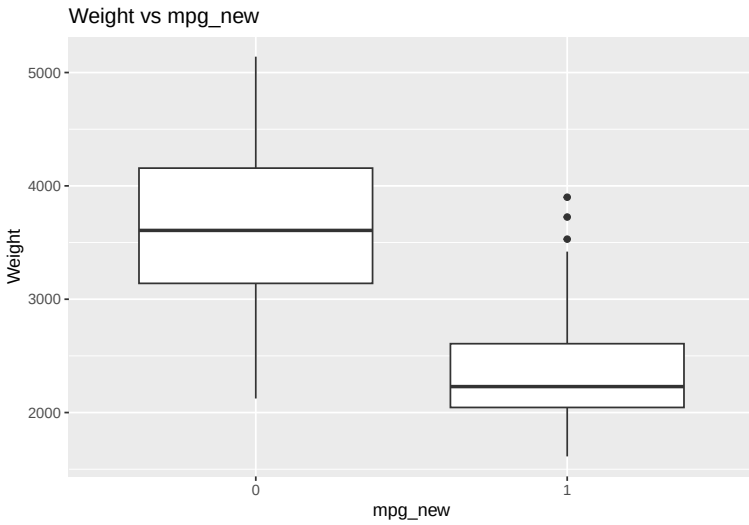
```
       x = "mpg_new",
       y = "Displacement")
```



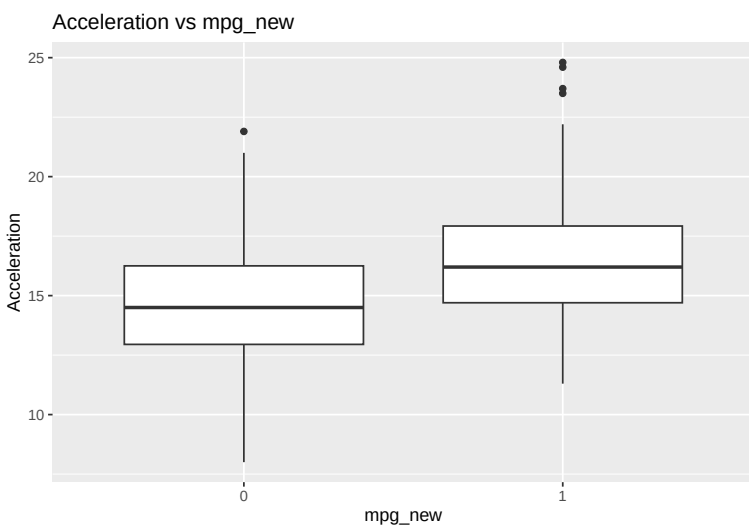

```
ggplot(Auto_2, aes(x = factor(mpg_new), y = horsepower)) +
  geom_boxplot() +
  labs(title = "Horsepower vs mpg_new",
       x = "mpg_new",
       y = "Horsepower")
```




```
ggplot(Auto_2, aes(x = factor(mpg_new), y = weight)) +
  geom_boxplot() +
  labs(title = "Weight vs mpg_new",
       x = "mpg_new",
       y = "Weight")
```




```
ggplot(Auto_2, aes(x = factor(mpg_new), y = acceleration)) +
  geom_boxplot() +
  labs(title = "Acceleration vs mpg_new",
       x = "mpg_new",
       y = "Acceleration")
```




#### Solution:

From the boxplots, `weight` appears to be most strongly associated with `mpg_new`, as there is a clear separation between cars with `mpg_new = 0` and `mpg_new = 1` and minimal overlap in their distributions. `Horsepower`, `displacement`, and `cylinders` also show strong differences between the two groups. `Acceleration` appears to have the weakest association, as the distributions for `mpg_new = 0` and `mpg_new= 1` overlap considerably.

## Part B

Using `weight` as the predictor, estimate a simple logistic regression model for the binary response `mpg_new`. Write the equation for your estimated model in *both* probability and log-odds form.

```
model <- glm(mpg_new ~ weight, data = Auto_2, family= "binomial")
summary(model)
```

```
##
## Call:
## glm(formula = mpg_new ~ weight, family = "binomial", data = Auto_2)
```

```
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) 11.763284   1.162411  10.120   <2e-16 ***
## weight      -0.004119   0.000413  -9.974   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 543.43  on 391  degrees of freedom
## Residual deviance: 228.31  on 390  degrees of freedom
## AIC: 232.31
##
## Number of Fisher Scoring iterations: 6
```

**Solution:** From the summary output, we have $\beta_0 = 11.763284$ and $\beta_1 = -0.004119$. The logistic regression model in log-odds form is: $\log\left(\frac{P(mpg_new=1|weight)}{1-P(mpg_new=1|weight)}\right) = \beta_0 + \beta_1 x = 11.763284 - 0.004119 * weight$. The probability form of the model is: $P(mpg_new = 1|weight) = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}} = \frac{1}{1+e^{-(11.763284-0.004119*weight)}}$.

Since the coefficient of weight is negative, as weight increases, the log-odds of having above-median mpg decreases. Therefore, heavier cars are less likely to have high fuel efficiency. For every 1-unit increase in weight, the log-odds of having above-median mpg decreases by 0.004119. In terms of odds: $e^{-0.004119} \approx 0.996$. This means each additional pound multiplies the odds of having high mpg by about 0.996.

### Part C

Show the six steps of a hypothesis test for the null hypothesis $H_0 : \beta_1 = 0$, where $\beta_1$ is the coefficient for the `weight` predictor in your model from Part B. What does the conclusion of this test mean in the context of a car's weight and fuel efficiency?

**Solution:** Here are the six steps of a hypothesis: - Define the null and alternate hypotheses - Choose a level of significance - Compute a test statistic - Construct a null distribution - Calculate a p-value - Decide whether to reject null hypothesis

Step1: Define the null and alternate hypothesis $H_0 : \beta_1 = 0$ $H_a : \beta_1 \neq 0$

Step2: Choose a level of significance: Let $\alpha = 0.05$.

Step3: Compute a test statistic: From the output $\beta_1 = -0.004119$ and $SE(\beta_1) = 0.000413$. Thus, z-statistic is: $z = \frac{\beta_1 - 0}{SE(\beta_1)} = \frac{-0.004119}{0.000413} \approx -9.974$.

Step 4: Construct a null distribution: Under $H_0 : \beta_1 = 0$, the test statistic approximately follows a standard normal distribution: $z$ $N(0, 1)$.

Step5: Calculate a p-value: Since the p-value is less than $2 * 10^{-16}$.

Step6: Decision and Conclusion: Since p-value is less than 0.05, we reject null hypothesis.

Overall, there is very strong evidence that a car's weight is associated with fuel efficiency. Because $\beta_1$ is negative, heavier cars are slightly less likely to have above-median mpg, meaning heavier cars tend to be less fuel efficient.

## Problem 4

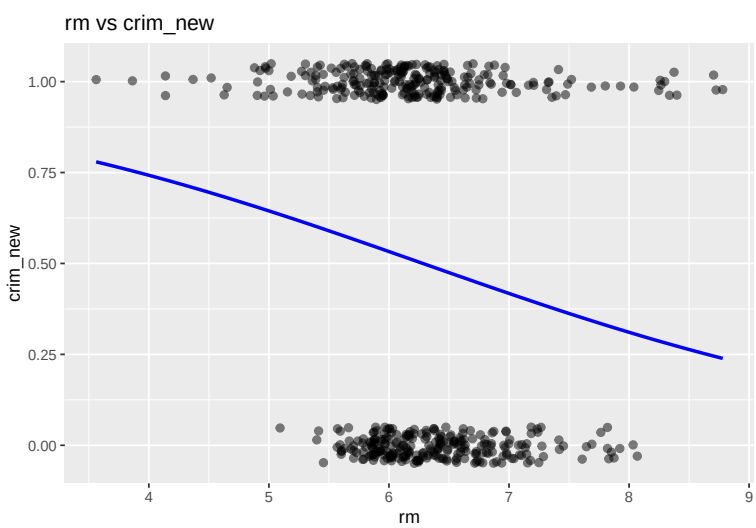
This problem uses the `Boston` data set from the `ISLR2` package.

```r
data(Boston)
```

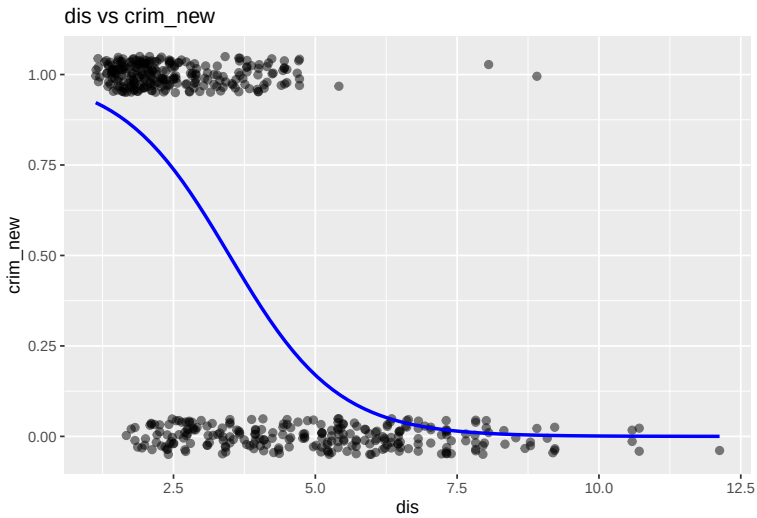
## Part A

Add a new binary variable `crim_new` that is equal to 1 when `crim` has a value greater than its median and 0 otherwise. Then, create four scatter plots with `crim_new` on the $y$-axis and `rm`, `dis`, `ptratio`, and `medv`, respectively, on the $x$-axis. Interpret the observed direction (positive or negative) of association in each plot. Does this make sense in the context of the data?

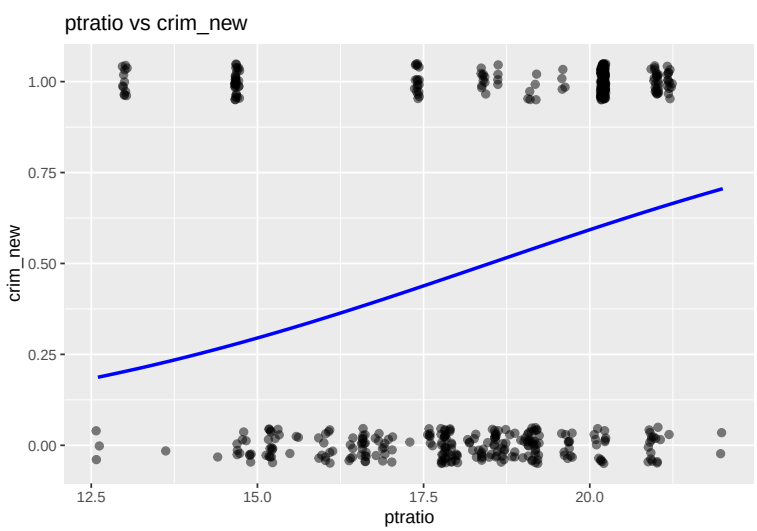
```r
crim_median <- median(Boston$crim, na.rm = TRUE)
Boston$crim_new <- ifelse(Boston$crim > crim_median, 1, 0)
ggplot(Boston, aes(x = rm, y =crim_new)) +
  geom_jitter(height = 0.05, size = 2, alpha = 0.5) +
  geom_smooth(
    formula = y ~ x, method = "glm",
    method.args = list(family = "binomial"), se = FALSE, color = "blue"
  )+
  labs(title = "rm vs crim_new",
       x = "rm",
       y = "crim_new")
```



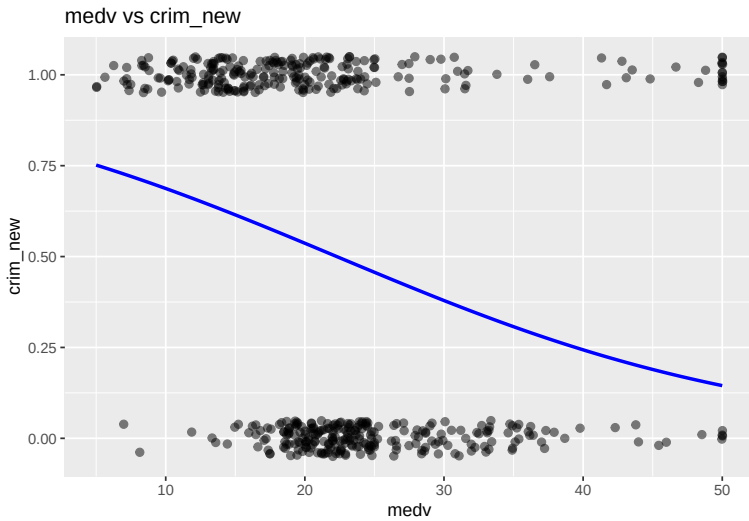

```r
ggplot(Boston, aes(x = dis, y =crim_new)) +
  geom_jitter(height = 0.05, size = 2, alpha = 0.5) +
  geom_smooth(formula = y~x, method = "glm",
              method.args = list(family = "binomial"), se = FALSE, color= 'blue') +
  labs(title = "dis vs crim_new",
       x = "dis",
       y = "crim_new")
```




```
ggplot(Boston, aes(x = ptratio, y = crim_new)) +
  geom_jitter(height = 0.05, size = 2, alpha = 0.5) +
  geom_smooth(formula = y~x, method = "glm",
              method.args = list(family = "binomial"), se = FALSE, color = 'blue') +
  labs(title = "ptratio vs crim_new",
       x = 'ptratio',
       y = 'crim_new')
```




```
ggplot(Boston, aes(x = medv, y = crim_new)) +
  geom_jitter(height = 0.05, size = 2, alpha = 0.5) +
  geom_smooth(formula= y~x, method = "glm",
              method.args = list(family = "binomial"), se = FALSE, color = 'blue') +
  labs(title = 'medv vs crim_new',
       x = 'medv',
       y = 'crim_new')
```




**Solution:** The logistic smooth curves show that `dis` has a strong negative association with `crim_new`, indicating that areas farther from employment centers tend to have lower crime rates. The variable `medv` also shows a strong negative relationship, suggesting that higher home values are associated with lower crime. The variable `ptratio` shows a positive association, meaning that higher pupil-teacher ratios are associated with higher crime. Finally, `rm` shows a moderate negative association, indicating that neighborhoods with more rooms tend to have lower crime. These relationships are consistent with socioeconomic expectations in the Boston data.

## Part B

Using `medv` as the predictor, estimate a simple logistic regression model for the binary response `crim_new`. Write the equation for your estimated model in *both* probability and log-odds form.

```r
boston_model <- glm(crim_new ~ medv, data = Boston, family = "binomial")
summary(boston_model)
```

```
##
## Call:
## glm(formula = crim_new ~ medv, family = "binomial", data = Boston)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.42773    0.26642   5.359 8.37e-08 ***
## medv        -0.06404    0.01141  -5.612 2.00e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 701.46  on 505  degrees of freedom
## Residual deviance: 664.48  on 504  degrees of freedom
## AIC: 668.48
##
## Number of Fisher Scoring iterations: 4
```

**Solution:** From the summary output: $\beta_0 = 1.42773$ and $\beta_1 = -0.06404$. The logistic regression model in log-odds form is: $\log\left(\frac{P(crim_new=1|medv)}{1-P(crim_new=1|medv)}\right) = \beta_0 + \beta_1 x = 1.42773 - 0.06404 * medv$. The probability form

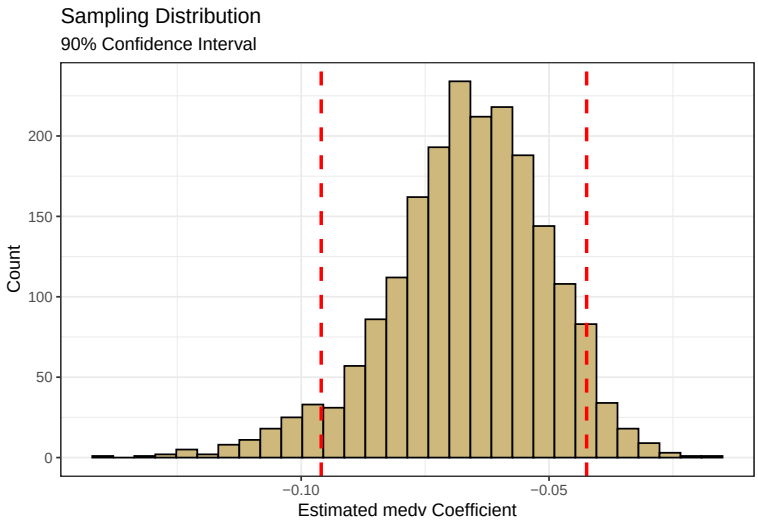
of the model is: $P(crim_new = 1|medv) = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}} = \frac{1}{1+e^{-(1.42773-0.06404*medv)}}$.

Since the coefficient of `medv` is negative, as median home value increases, the log-odds of having above-median crime decreases. Therefore, higher home values are associated with lower crime rates. More specifically, for each one-unit increase in `medv`, the log-odds of high crime decreases by 0.06404. In terms of odds: $e^{-0.06404} \approx 0.938$. This means each additional unit increase in median home value multiplies the odds of high crime by about 0.938.

## Part C

Construct a 90% bootstrap confidence interval for the parameter $\beta_1$, where $\beta_1$ is the coefficient for the `medv` predictor in your model from Part B. What do the bounds on this confidence interval suggest in terms of the association between home values and crime rates?

```r
set.seed(1)

results2 <- data.frame(beta1 = rep(NA, 2000))

for (i in 1:2000) {
  boot <- Boston[sample(nrow(Boston), replace = TRUE), ]

  remodel2 <- glm(crim_new ~ medv, data = boot, family = "binomial")
  results2$beta1[i] <- coef(remodel2)[2]
}

threshold1 <- as.numeric(quantile(results2$beta1, 0.05))
threshold2 <- as.numeric(quantile(results2$beta1, 0.95))

ggplot(data = results2, aes(x = beta1)) +
  geom_histogram(color = "black", fill = "#CFB87C") +
  geom_vline(
    xintercept = threshold1, color = "red",
    linewidth = 1, linetype = "dashed"
  ) +
  geom_vline(
    xintercept = threshold2, color = "red",
    linewidth = 1, linetype = "dashed"
  ) +
  labs(
    title = "Sampling Distribution",
    subtitle = "90% Confidence Interval",
    x = "Estimated medv Coefficient",
    y = "Count"
  ) +
  theme_bw()
```




```r
quantile(results2$beta1, probs = c(0.05, 0.95))
```

```
##          5%         95%
## -0.09596346 -0.04241499
```

**Solution:** A 90% bootstrap confidence interval for $\beta_1$ is approximately (-0.09596, -0.04241). Since the entire interval is negative and does not contain 0, this suggests a statistically significant negative association between median home value and crime rate. Specifically, as median home value increases, the probability of having above-median crime decreases.